

$$\bar{i}_1 = \bar{i}_4 + \bar{i}_3 \rightarrow \frac{V_i - V_x}{R_1} = \frac{V_x}{R_4} + \frac{V_x}{R_3}$$

$$\left(R_3 + R_4 + \frac{R_3 R_4}{R_1} \right) V_x = \frac{R_3 R_4}{R_1} V_i$$

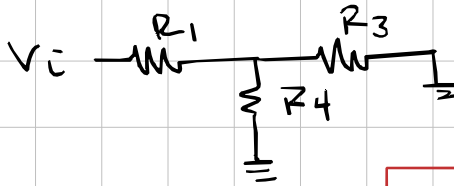
$$V_x = R_3 R_4 V_i / (R_1 R_3 + R_1 R_4 + R_3 R_4)$$

$$\bar{i}_3 = \bar{i}_2$$

$$\frac{V_x}{R_3} = \frac{0 - V_o}{R_2} \rightarrow \frac{R_4 V_i}{R_1 R_3 + R_1 R_4 + R_3 R_4} = -\frac{V_o}{R_2}$$

$$A_{ideal} = -R_2 R_4 / (R_1 R_3 + R_1 R_4 + R_3 R_4)$$

b) Ideal $R_i \rightarrow$ virtual short still exists:



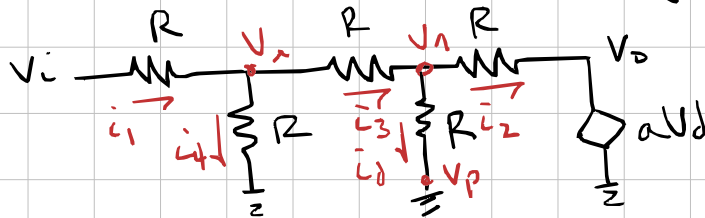
$$R_i = R_1 + R_3 \parallel R_4$$

$$R_i = R_1 + R_3 R_4 / (R_3 + R_4)$$

$$r_d = R, r_o \neq, a \neq \infty, R_1, R_2, R_3, R_4 = R$$

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c) $r_o = 0$, find A via exact analysis



$$V_o = aV_d \\ \rightarrow V_n = -V_o/a$$

$$i_1 = i_3 + i_4 \rightarrow \frac{V_i - V_x}{R} = \frac{V_x}{R} + \frac{V_x - V_n}{R}$$

$$V_i - V_x = V_x + V_x - V_n \rightarrow V_i = 3V_x - V_n \quad \textcircled{a}$$

$$i_3 = i_1 + i_2 \rightarrow \frac{V_x - V_n}{R} = \frac{V_n}{R} + \frac{V_n - V_o}{R}$$

$$V_x - V_n = V_n + V_n - V_o \rightarrow V_x = 3V_n - V_o \quad \textcircled{b}$$

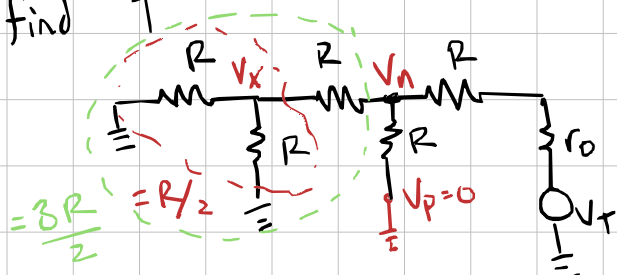
$$\textcircled{b} \rightarrow \textcircled{a} \quad V_i = 3(3V_n - V_o) - V_n = 8V_n - 3V_o$$

$$V_n = -V_o/a \rightarrow V_i = -\frac{8V_o}{a} - 3V_o = -V_o \left(\frac{8}{a} + 3 \right)$$

$$V_o = -1/3 + 8/a = -\frac{1}{3} / 1 + 8/3a$$

e) find T

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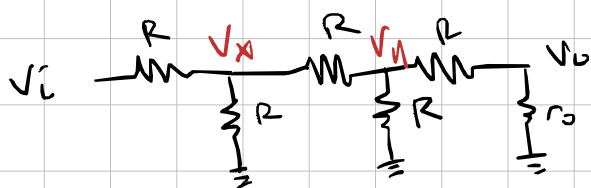
$$V_n = V_T \frac{3R/2 \parallel R}{3R/2 \parallel R + R + r_o} \quad \frac{3R/2 \parallel R}{3R/2 \parallel R + R + r_o} = \frac{\frac{3}{2}R^2}{\frac{3}{2}R + R} = \frac{3/2R}{5/2} = \frac{3}{5}R$$

$$\frac{V_n}{V_T} = \frac{3}{5}R / \left(\frac{3}{5}R + R + r_o \right) = \frac{3}{5}R / \left(\frac{8}{5}R + r_o \right) = 3R / (8R + 5r_o)$$

$$T = -a_{v0}/V_T, \quad V_d = -V_n \rightarrow -\frac{V_d}{V_T} = \frac{V_n}{V_T}$$

$$T = 3Ra / (8R + 5r_o)$$

f) find a_{ft}



(start with same analysis as (c))

$$\text{KCL @ } V_x: \frac{V_i - V_x}{R} = \frac{V_x - V_n}{R} + \frac{V_x}{R}$$

$$V_i - V_x = V_x - V_n + V_x \rightarrow V_i = 3V_x - V_n \quad \textcircled{a}$$

$$\text{KCL @ } V_n: \frac{V_x - V_n}{R} = \frac{V_n}{R} + \frac{V_n}{R + r_o}$$

$$\frac{V_x - V_n}{R} = \frac{V_n}{R} + \frac{V_n - V_o}{R}$$

$$V_x - V_n = V_n + V_n - V_o \rightarrow V_x = 3V_n - V_o \quad \textcircled{b}$$

$$\textcircled{b} \rightarrow \textcircled{a} \quad V_i = 3(3V_n - V_o) - V_n = 8V_n - 3V_o$$

$$V_o = V_n \frac{r_o}{R + r_o} \quad \checkmark$$

$$V_i = 8 \left[\frac{R + r_o}{r_o} \right] V_o - 3V_o = V_o \left[\frac{8R + 5r_o}{r_o} \right]$$

$$a_{ft} = r_o / (8R + 5r_o)$$

$$g) \quad A = A_{ideal} / (1 + 1/T) + a_{ft} / (1 + T)$$

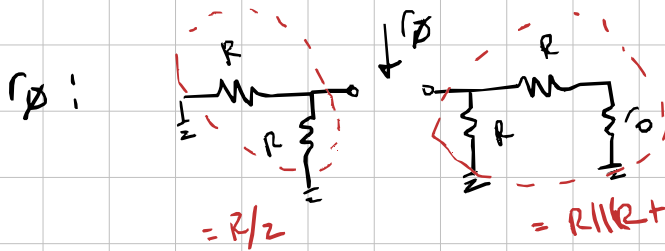
$$A_{ideal} = -R_2 R_4 / (R_1 R_3 + R_1 R_4 + R_2) \\ A_{ideal} = -1/3 \quad R_{1,2,3,4} = R$$

$$A = -\frac{1}{3} / \left(1 + \frac{8R + 5r_o}{3aR} \right) + \frac{r_o / (8R + 5r_o)}{1 + 3aR / (8R + 5r_o)}$$

validating with (c): set $r_o = 0$

$$A = -\frac{1}{3} / \left(1 + \frac{8R}{3aR} \right) + 0 = -\frac{1}{3} / \left(1 + \frac{8}{3a} \right) \quad \checkmark$$

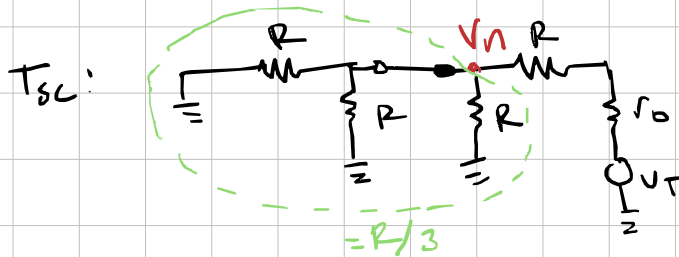
h) find R_{c3} (R_3) solution uses Blackman's formula, could do Directly...



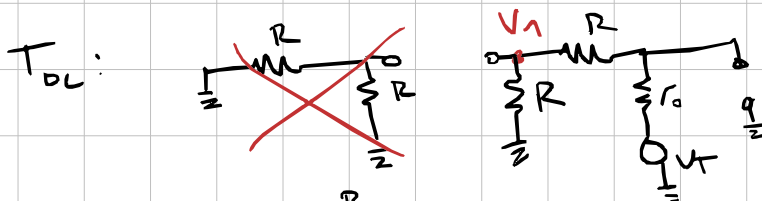
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$$r_\phi = R/2 + \frac{R(R+r_o)}{2R+r_o} = \frac{R(2R+r_o)}{2(2R+r_o)} + \frac{2R(R+r_o)}{2(2R+r_o)}$$

$$r_\phi = \frac{4R^2 + 3Rr_o}{2(2R+r_o)} = \frac{R(4R+3r_o)}{2(2R+r_o)}$$



$$v_1 = v_T \frac{R/3}{R/3 + R + r_o} = v_T \frac{R}{4R+3r_o} \rightarrow T_{sc} = \frac{aR}{4R+3r_o}$$



$$v_1 = v_T \frac{R}{2R+r_o} \rightarrow T_{oc} = \frac{aR}{2R+r_o}$$

$$R_{c3} = \frac{R(4R+3r_o)}{2(2R+r_o)} \left[\frac{1 + aR/4R+3r_o}{1 + aR/2R+r_o} \right]$$

$$R_{c3} = \frac{R}{2} \left[\frac{4R+3r_o+aR}{2R+r_o+aR} \right]$$

i) find a_2 . Note: a_2 implies $r_0 = 0$ otherwise we would have T!

→ want $L = a_2 b$

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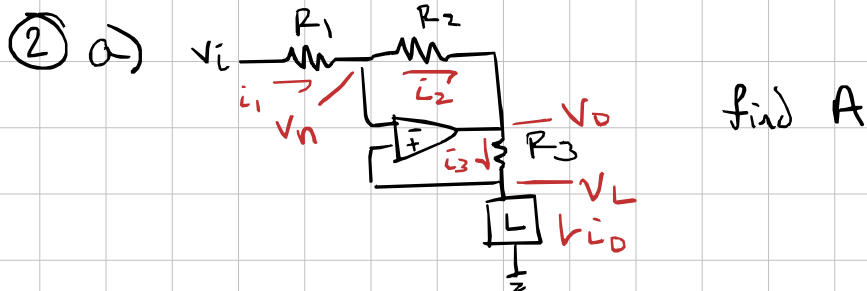
from c) or g) with $r_0 = 0$

$$A = -\frac{1}{3} / 1 + 8/3a \quad \therefore L = 3a/8$$

$$A_{ideal} = -\frac{1}{3} \rightarrow b = -3$$

$$L = a_2 b \Rightarrow \frac{3a}{8} = a_2 (-3)$$

$$a_2 = -a/8$$



By virtual short: $V_n = V_L$

$$\bar{i}_3 = \bar{i}_0 = \frac{V_o - V_L}{R_3} \quad \bar{i}_1 = \bar{i}_2 \Rightarrow \frac{V_i - V_L}{R_1} = \frac{V_L - V_o}{R_2}$$



$$V_o - V_L = \frac{R_2}{R_1} (V_L - V_i)$$

$$\bar{i}_0 = \frac{R_2}{R_1 R_3} (V_L - V_i)$$

$$A = -\frac{R_2}{R_1 R_3} \quad R_o = -\frac{R_1 R_3}{R_2}$$

b) $V_L = 2V$, $\bar{i}_0 = 4mA$, $R_1, R_2, R_3 = 1k\Omega$, $\bar{i}_3 = 2mA$, $V_S = \pm 10V$

find V_o & $\bar{i}$ out of amplifier $\bar{i}_0'$

$$\bar{i}_0 = \frac{V_o - V_L}{R_3} \rightarrow V_o = (4mA)(1k) + V_L = 6V$$

$$\bar{i}_0' = -\bar{i}_2 + \bar{i}_3 = \frac{V_o - V_L}{R_2} + \frac{V_o - V_L}{R_3} = 2\bar{i}_0 = 8mA$$

$$P_{out} = (10V - 6V)(8mA) = 32mW$$

$$P_g = (10V - (-10V))(2mA) = 40mW$$

$$P = 72mW$$

③ $A = 100 \text{ V/V}$, $\text{CMRR} = 80 \text{ dB}$

$$V_1 = 2.47 \text{ V}$$

$$V_2 = 2.56 \text{ V}$$

Since CMRR fairly large, know: $A_{DM} \approx A$
and $A_{CM} \ll A_{DM}$

$$\text{CMRR} = 80 \text{ dB} = 10000 = \frac{A_{DM}}{A_{CM}} = \frac{100}{A_{CM}}$$
$$A_{CM} = 0.01$$

$$V_{out} = A_{DM} V_{DM} + \underbrace{A_{CM} V_{CM}}_{\text{error}}$$

$$A_{CM} V_{CM} = 0.01 (2.56 + 2.47) / 2$$

$$\text{error} = 25.2 \text{ mV} \rightarrow 2.8\%$$

for reference, $i_{\text{ref}} = 100 (2.56 - 2.47) = 9 \text{ V}$

Actual output: 9.0252 V